

PFAS Release Mapping of Articles Used in Semiconductor Manufacturing

Semiconductor PFAS Consortium Articles Working Group

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The Semiconductor PFAS Consortium is an international group of semiconductor industry stakeholders formed to collect the technical data needed to formulate an industry approach to perfluoroalkyl and polyfluoroalkyl substances (PFAS). Consortium membership comprises semiconductor manufacturers and members of the supply chain, including chemical, material and equipment suppliers.

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Abstract

Articles used in semiconductor manufacturing represent a diverse collection of parts, components, consumables, assemblages and subsystems, loosely grouped together to represent a class of essential items distinct from the chemicals, gases and materials that are also required. Tubing, piping, filters, drums, vessels, valves, storage boxes, seals, wafer handlers and other equipment components represent just a small fraction of the objects classifiable as articles.

Given demanding specifications for purity, cleanliness and reliability, coupled with extreme temperatures and contact with aggressive chemistries such as strong acids, bases, oxidizers and plasmas, articles must be able to withstand harsh conditions, have long durability, not contribute to defectivity, and not leach material from or into the contacted materials being delivered, measured or processed.

The properties of PFAS-containing materials include thermal and chemical resistance, extended usable life, flexibility and lubricity, and manufacturability. This class of materials is undergoing increasing levels of scrutiny, as some small-molecule PFAS-containing materials have been shown to have undesirable health effects.

Understanding the potential release of PFAS-containing articles to the environment requires comprehending the different types of articles, their role in the semiconductor manufacturing process, the potential release points during use, and how manufacturers dispose of them at their end of life. Articles range from those with a relatively short life span – ranging from single use to several months' duration – to those that are essentially permanent components of a semiconductor manufacturing facility and may have a useful lifetime of 25 years or more.

Mapping the sources and release pathways of PFAS-containing articles throughout the life cycle of a semiconductor manufacturing facility reveals the complexities and diversity of potential releases. Numerous examples of different PFAS-containing articles and their potential release modes further illustrate the ubiquitous nature of these items and their importance in semiconductor manufacturing.

Background

Per- and poly-fluoroalkyl substances (PFAS) are manmade chemicals used in many everyday applications with specific beneficial properties such as chemical stability. Recently there is suggestion that members of this class of compounds exhibit high environmental persistence (vP), which is combined in some members with bioaccumulation (B) and toxicity (T). It is highly probable that PFAS chemicals will be subject to significant regulatory restrictions in the future. At the same time, PFAS-containing materials are crucial components used in semiconductor manufacturing, to the extent that the manufacture of advanced semiconductor chips is currently not possible without them.

The Semiconductor PFAS Consortium has chosen the broadest definition found in recent proposed regulations and includes in its definition of PFAS all chemistries and materials that contain molecules with -CF₂- and/or -CF₃ groups. This definition closely aligns with the Organization for Economic Cooperation and Development definition¹. Despite the consortium's broad definition of PFAS, not all PFAS-containing materials within the definition are considered bioaccumulative or toxic.

The fluorinated organic molecule in the form of a fluoropolymer or fluoroelastomer possesses numerous attributes that provide unique functionality across a wide spectrum of applications. In many situations, it is not just one particular attribute (such as low surface tension) that makes a fluoropolymer or fluoroelastomer effective for an application, but the combination of several attributes that enable fluorinated organics to satisfy multiple, overlapping performance requirements.

Given the importance of fluorinated organic chemicals in semiconductor manufacturing, the Semiconductor PFAS Consortium has developed seven white papers², three case studies and a paper detailing the “Impact of a Potential PFAS Restriction on the Semiconductor Sector.”³ These papers identify PFAS uses in semiconductor manufacturing and assess where their use meets the concept of “essential.”⁴ In their papers, the authors conclude that for most PFAS use cases, non-PFAS alternatives providing the same performance characteristics are currently unavailable.

Figure 1 is a broad overview of the scope of the Semiconductor PFAS Consortium’s analysis, showing both front- and back-end semiconductor processing, as well as facility support functions and device assembly, test and packaging operations. The scope of the operations considered by the Semiconductor PFAS Consortium starts with a bare silicon wafer entering a manufacturing facility and ends at the packaging of semiconductor devices, while including semiconductor and manufacturing related equipment (SMRE) and the facility infrastructure required to manufacture semiconductor devices. The scope does not include the manufacturing of articles, SMRE or facility infrastructure components, which can be potential sources of release in their own right. Similarly, the scope does not include potential releases from the use of products generated by semiconductor manufacturing or packaging processes.

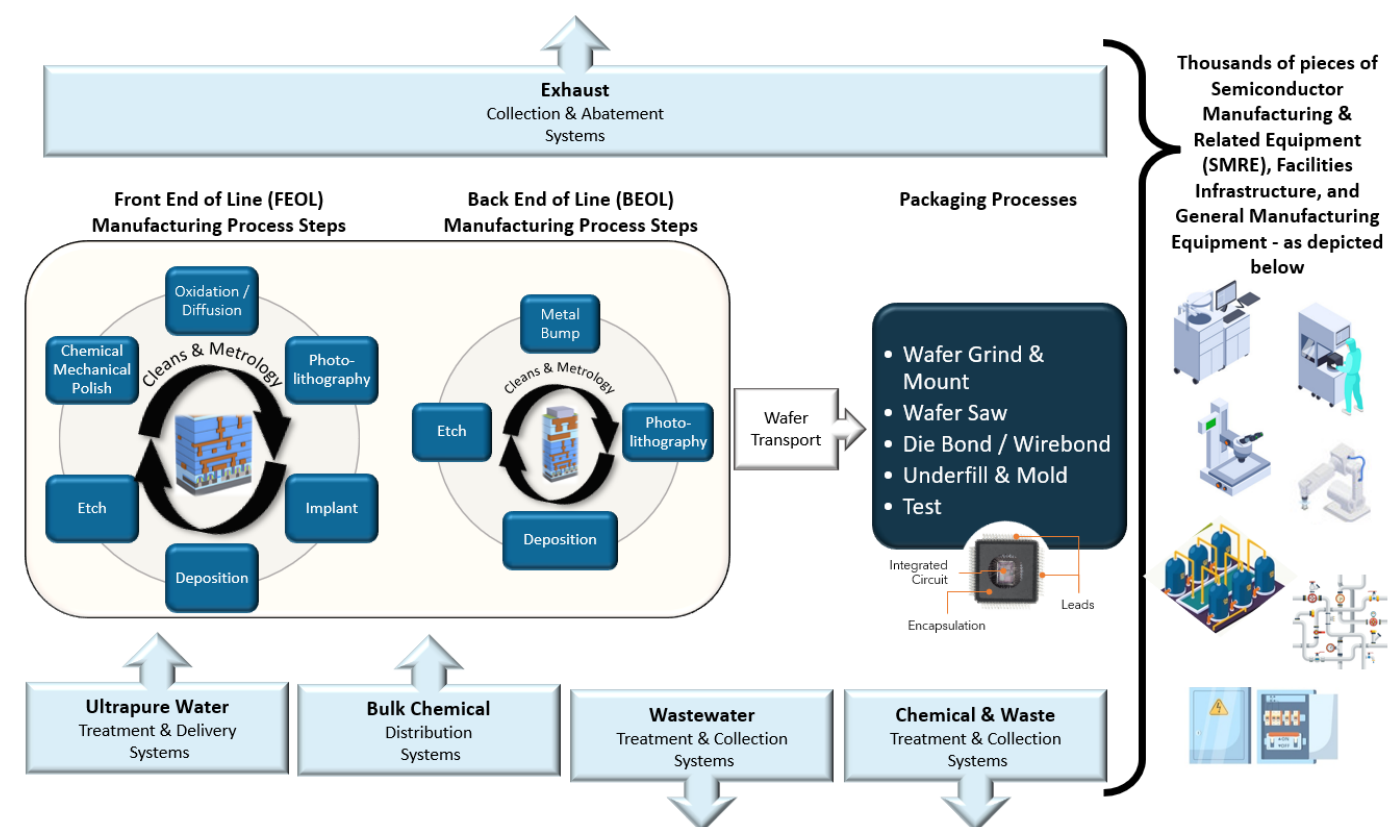


Figure 1: General overview of semiconductor manufacturing process steps, operations and systems evaluated by the Semiconductor PFAS Consortium.

Objective

The objective of this paper is to identify the principal, PFAS environmental release pathways from the use of articles in semiconductor manufacturing facilities and provide a generalized template for the development of specific PFAS release quantification models for semiconductor manufacturing.

Conceptual Release Map

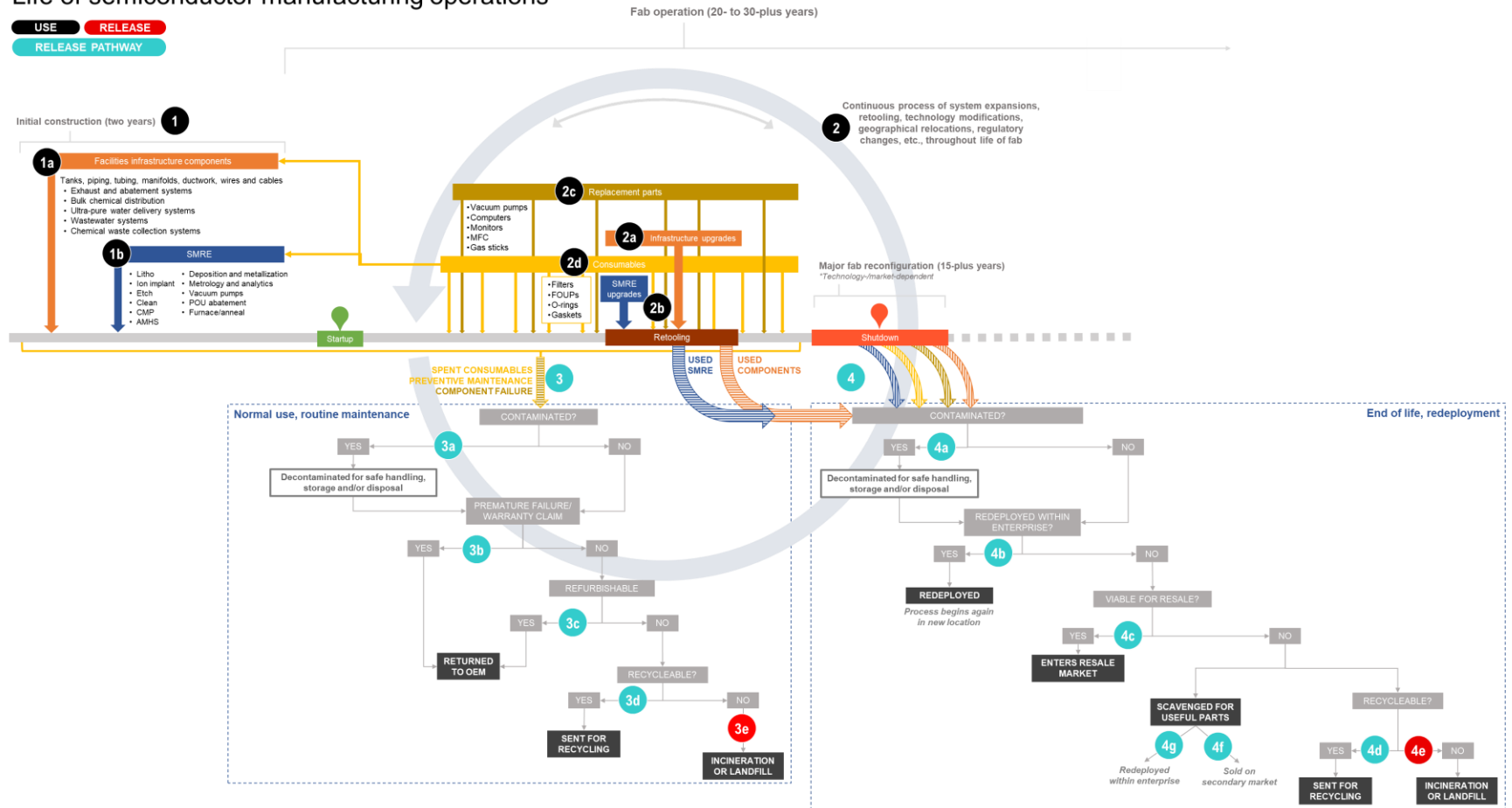
Figure 2 shows a schematic representation of the life cycle of articles: their introduction, use, retire and disposal from a semiconductor manufacturing facility. Numbers indicate various means of use (in black), release pathways (in blue) or release (red).

Initial facility bring-up (1) may introduce an article in both the facilities infrastructure (1a) and the SMRE (1b). During facility operation (2), adding a new infrastructure (2a) or SMRE (2b) may include the introduction of new articles. Regular equipment and facilities use drives the consumption of replacement parts (2c) and consumables (2d). Articles consumed or replaced exit the facility (3), and if contaminated with process chemicals or residues, may undergo decontamination (3a).

Premature use or failure may cause the return of parts or consumables to the supplier (3b). If the spent article is eligible for reuse, it may be refurbished (3c) or otherwise recycled (3d). If these options are not viable, the manufacturer may dispose of the article through incineration or send it to a solid waste landfill (3e). Similarly, Figure 2 shows paths for the retirement of parts at end of life or shutdown (4, 4a-e).

Table 1 details each example illustrated in Figure 2 and includes estimates of the duration of use (lifetime) of each release map variable. Figure 2 is representative of a semiconductor fabrication facility (fab), but is additionally representative of other semiconductor manufacturing facilities (such as photomask and assembly test and packaging), which have analogous infrastructures, SMRE and use models, and spare parts and consumables models.

Life of semiconductor manufacturing operations*



*Representative overview; inclusive of chip manufacturing, mask manufacturing, assembly, test, and packaging. Does not include upstream manufacturing processes for articles or materials.

Figure 2: Representative life of articles in semiconductor manufacturing operations and facility infrastructure.

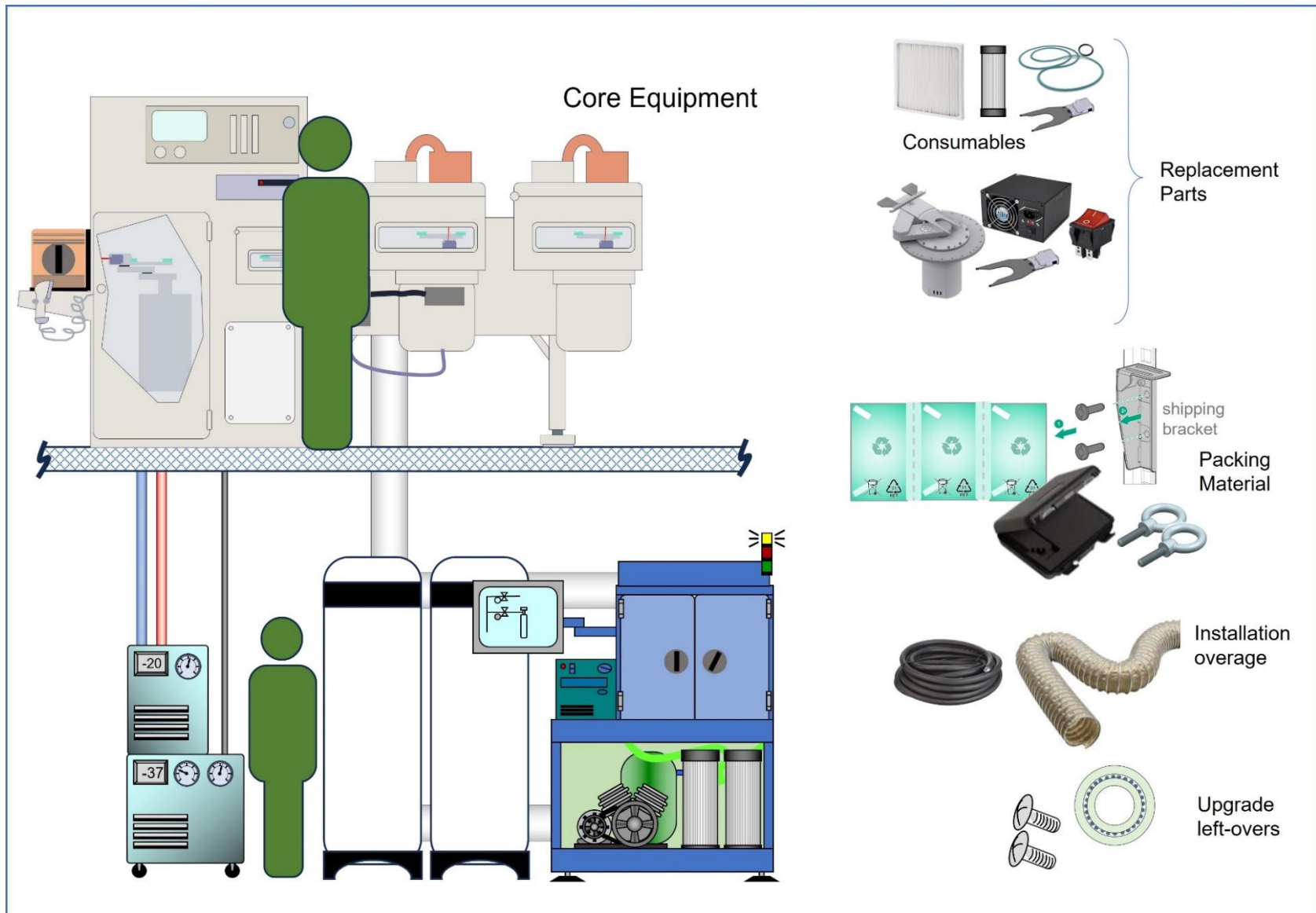


Figure 3: Examples of SMRE core equipment.

Table 1: Potential PFAS-containing article time in service and release pathways.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
1	Initial facility construction		
1a	Installation of facilities infrastructure and components	25 plus years	Generally, a factory has a greater than 25-year life span, as do facility systems. Bulk chemical distribution systems may have a shorter life span because of process chemistry changes; thus, construction materials have a 25-year useful life span.
1b	Installation of SMRE	25 plus years The lifetime of individual pieces of equipment can vary	A modern semiconductor manufacturing facility contains hundreds of pieces of processing equipment. Fluoropolymers and fluoroelastomers play a pivotal role in semiconductor manufacturing equipment given their exceptional chemical resistance, thermal stability and low particle generation properties. Additionally, some articles are made of a non-PFAS polymer that contains PFAS from processing aids or additives. Manufacturers use articles within SMRE in various ways, depending upon the article and its supply chain. They purchase some articles for direct use, such as perfluoroalkoxy copolymer (PFA) tubing cut to length from larger spools. Other articles, such as a flow controller, are contained within a part purchased from other suppliers for either direct use or as part of an assembly. Finally, parts and assemblies may be combined into additional assemblies and integrated into higher-level equipment before being supplied to the facility.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
2	Normal facility operations		Includes routine maintenance, retooling and process modifications
2a	Installation/major reconfiguration of facilities infrastructure and components	15-plus years	If a factory has been repurposed for another process or function (like C4 bump back end), there may be a need to rebuild some facility systems to accommodate new process requirements or equipment. Old facility materials may not be compatible with new processes or not required, necessitating their removal or replacement. This is more likely for bulk chemical delivery systems and waste systems. Occasionally, facilities may have entire production lines reconfigured or completely shut down. During construction excess material (overage) deemed unusable may be discarded.
2b	Installation/upgrade of SMRE	15-plus years	At various times during the lifetime of a semiconductor manufacturing facility, existing equipment may require modification or replacement because of changes in the chip design or the migration of a process from another facility. Occasionally, major upgrades are provided for equipment. During equipment or infrastructure installation excess material (overage) deemed unusable may be discarded.
2c	Replacement parts	Various, as needed	Articles contained in SMRE and facilities infrastructure may need replacing when failing prematurely, or when newer articles become available with improved performance characteristics. The equipment supplier evaluates new or changed replacement parts for proper performance before using them within the original equipment supplier.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
2d	Consumables	Weekly to one year	Manufacturers typically replace certain articles (filters, gaskets, O-rings) on a regular preventive maintenance schedule of varying time periods. Consumables are a subset of replacement parts are replaced per a preventive maintenance to keep processing equipment operating properly and avoid unwanted downtime caused by an unexpected component failure.
3	Spent consumables, preventive maintenance, component failure		
3a	Removal of spent consumables or components from preventive maintenance or component failure to determine their potential contamination	Variable	Manufacturers evaluate spent consumables or end-of life components to determine whether they are contaminated from process or facilities chemistries. If the article is contaminated, it may be decontaminated for safe handling, storage or disposal.
3b	Decision point for premature failure/warranty claim	Variable	Manufacturers return the article to the supplier if it fails prematurely or as part of a warranty claim. Once returned, the supplier typically repairs or replaces the article, with any waste generated being their responsibility.
3c	Decision point on the possibility of refurbishing the article	Variable	The article may be refurbished on-site, at a supplier location or at the original manufacturer, with waste generated from the refurbishment process managed according to the waste management hierarchy. The manufacturer would then return the article to service or place it in a parts crib for future use. If it is not possible to refurbish the article, it moves to the next step of the life-cycle flow diagram.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
3d	Spent consumables and end-of-life components sent for recycling	Variable	Articles no longer suitable for manufacturing or operations use that are sent for recycling may include high-value fluoropolymers such as PFA or polytetrafluoroethylene (PTFE), or articles primarily made of other recyclable materials (such as metals) that are coated with or contain PFAS components.
3e	Spent consumables and end-of-life components sent for incineration or landfill disposal	Variable	Manufacturers send articles no longer suitable for manufacturing or operations use that cannot be sent for recycling (such as fluoropolymers that do not have a recycling market or cannot be decontaminated) for incineration or disposal in a landfill. The decision for incineration vs. landfill disposal most often depends on local regulations. In Europe, these articles would be burned in a waste-to-energy facility. In the U.S., they are typically sent to a municipal solid waste landfill unless classified as regulated waste, in which case they would be sent to a permitted hazardous waste landfill.
4	End of life, asset redeployment		
4a	Removal of SMRE, components or facilities infrastructures that are at their end of life or no longer used, and may be redeployed	Variable	Manufacturers evaluate SMRE, components or facilities infrastructures to determine whether they are contaminated from process or facilities chemistries. If the article is contaminated, they decide whether to decontaminate the article for safe handling, storage or disposal. In many cases, it is possible to decontaminate SMRE. In other cases, it is difficult to remove certain process chemistries, or they have become embedded in the article.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
4b	SMRE, components or facilities infrastructures that are redeployed	Variable	SMRE, components or facilities infrastructures that still have a useful life can be deployed within the facility or at another company location. There may be some waste generated during redeployment if components need replacement to maintain functionality or to fit a new use, with any waste generated managed according to the waste management hierarchy.
4c	SMRE, components or facilities infrastructures that may go into the resale market	Variable	SMRE, components or facilities infrastructures that still have a useful life can enter the resale market. There may be some waste generated during resale if some components need replacement to maintain functionality or to fit a new use, with any waste generated managed according to the waste management hierarchy.
4d	SMRE, components or facilities infrastructures that may be recycled	Variable	SMRE, components or facilities infrastructures that are no longer suitable for manufacturing or operations use can be sent for recycling. These may include high-value fluoropolymers such as PFA or PTFE, or articles primarily made of other recyclable materials (such as metals) that are coated with or contain PFAS components. PFAS materials commingled with other recyclable materials are not in the scope of this assessment.
4e	SMRE, components or facilities infrastructures that cannot be recycled and must be disposed	Variable	Manufacturers send PFAS-containing articles that do not have a recycling market for incineration or disposal in a landfill. The decision for incineration vs. landfill disposal most often depends on local regulations. In Europe, these articles would be burned in a waste-to-energy facility. In the U.S., they are typically sent to a municipal solid waste landfill unless classified as regulated waste, in which case they would be sent to a permitted hazardous waste landfill.

Release Map Variable	Release Map Variable Description	Time in Service/ Life Cycle	Comments
4f	SMRE, components or facilities infrastructure articles may be scavenged for valuable parts that may be resold	Variable	SMRE, components or facilities infrastructures that are not viable for resale may be disassembled and individual parts may be resold onto the secondary market for future use. The scavenging process may generate waste, which the manufacturer would then manage according to the waste management hierarchy. If it is not possible to resell the scavenged part outside of the enterprise, it moves to the next step of the life-cycle flow diagram (4d/4e).
4g	SMRE, components or facilities infrastructure articles may be scavenged for valuable parts for reuse by the manufacturer	Variable	SMRE, components or facilities infrastructures that are not viable for resale may be disassembled and individual parts may be scavenged and retained in parts inventory for future use by the enterprise. The scavenging process may generate waste, which the manufacturer would then manage according to the waste management hierarchy. If it is not possible to retain the article/part for future use, it moves to the next step of the life-cycle flow diagram (4d/4e).

Release factors and data sources are listed as “undetermined” if industry data has not been collected to support quantification. Numeric release factors may be evaluated during Consortium quantification work

Conclusions and Next Steps

As shown in the process mapping and enumeration of the use of articles in chip manufacturing processes, identifying sources of potential release is highly involved. The next step in this process is to estimate the quantification of the release in order to identify the largest potential sources and mechanisms.

The industry, through collaborative engagements like the Semiconductor PFAS Consortium, are engaging in detailed studies of the propensity for articles in the manufacturing process to release PFAS-containing materials during their regular use. In parallel, the semiconductor industry continues to actively identify, test and implement improved process controls that minimize releases to the environment, including PFAS-containing materials. There is also an increasing focus on developing circularity for PFAS-containing materials, including fluoropolymers, through reuse and recycling.

Finally, the industry is and has been researching prospective PFAS substitutes. The identification and validation of alternatives is an expensive and lengthy process that entails identifying potential substitutes, evaluating their potential human and environmental risks, conducting laboratory and pilot tests, designing and retooling equipment, and conducting verification tests and qualification in a high-volume manufacturing environment before final implementation.

By taking a holistic, multifaceted approach to identify and minimize potential releases, seek circularity solutions, and explore alternative materials, the semiconductor articles community is continuing to demonstrate its commitment to being good stewards of this unique class of materials.

¹ OECD, Organization for Economic Cooperation and Development. 2021. *Reconciling Terminology of the Universe of Per- and Polyfluoroalkyl Substances: Recommendations and Practical Guidance*. OECD Series on Risk Management.

² Semiconductor PFAS Consortium technical white papers and case studies can be found at <https://semiconductors.org/pfas>.

³ RINA. 2023. *The Impact of a Potential PFAS Restriction on the Semiconductor Sector*. Impact Assessment, Leatherhead: RINA Tech UK Ltd.

⁴ Cousins, Ian T., Gretta Goldenman, Dorte Herzke, Rainer Lohmann, Mark Miller, Carla A. Ng, Sharyle Patton et al. 2019. "The concept of essential use for determining when uses of PFASs can be phased out." *Environmental Science: Processes & Impacts* 1803-1815.